

TOPOLOGY

SAATHVIK BOGAM

A Self Study

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1 Spaces, Balls and Openness

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DEFINITION 1.1 (Metric Space). Let X be a non empty set, and $d : X \times X \rightarrow \mathbb{R}$ be a metric on X such that $\forall x, y, z \in X$:

1. $d(x, y) \geq 0$, and $d(x, y) = 0 \iff x = y$
2. $d(x, y) = d(y, x)$
3. $d(x, z) + d(z, y) \geq d(x, y)$

Then the pair (X, d) is called a *metric space*.

DEFINITION 1.2 (Inner Product Space). If V is a real vector space, and $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called an inner product if $\forall x, y, z \in V$:

1. $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0 \iff x = 0$
2. $\langle x, y \rangle = \langle y, x \rangle$
3. $\langle x + z, y \rangle = \langle x, y \rangle + \langle z, y \rangle$
4. $\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$ where $\alpha \in \mathbb{R}$

Then $(V, \langle \cdot, \cdot \rangle)$ is called an *inner product space*.

DEFINITION 1.3 (Norm Linear Space). Let V be a real vector space and $\|x\|$ be the norm of a vector $x \in V$, then $(V, \|\cdot\|)$ is a norm linear space if $\forall x, y \in V$:

1. $\|x\| \geq 0$ and $\|x\| = 0 \iff x = 0$.
2. $\|ax\| = |a| \|x\|$ where $a \in \mathbb{R}$.
3. $\|x + y\| \leq \|x\| + \|y\|$.

THEOREM 1.4 (Cauchy–Schwarz Inequality). Let V be an inner product space and $\|x\| = \sqrt{\langle x, x \rangle}$ be the norm of $x \in V$, then for $x, y \in V$ we have:

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad (1)$$

Proof. First, we assume $\|x\| = \|y\| = 1$, then consider $\langle x - y, x - y \rangle$:

$$\langle x - y, x - y \rangle = \langle x, x \rangle + \langle y, y \rangle - 2\langle x, y \rangle \quad (2)$$

$$= 2(1 - \langle x, y \rangle) \geq 0 \implies \langle x, y \rangle \leq 1 \quad (3)$$

Similarly, for $\langle x + y, x + y \rangle$ we get $\langle x, y \rangle \geq -1$. Hence,

$$|\langle x, y \rangle| \leq 1 = \|x\| \|y\| \quad (4)$$

Now, for the general non-zero case, let $u = x/\|x\|$ and $v = y/\|y\|$, then $\|u\| = \|v\| = 1$ and $\langle u, v \rangle \leq 1$, by the previous case.

$$\left\langle \frac{x}{\|x\|}, \frac{y}{\|y\|} \right\rangle \leq 1 \implies \langle x, y \rangle \leq \|x\| \|y\| \quad (5)$$

The case for $\|x\| = 0$ or $\|y\| = 0$ are trivial. $\square$

DEFINITION 1.5. Let V be a real vector space and $x, y, z \in V$. Then z lies between x and y $\iff \exists t \in [0, 1]$ such that $z = tx + (1 - t)y$.

THEOREM 1.6. Let $x, y \in V$, where V is an inner product space with the norm metric. Then $d(x, y) = d(x, z) + d(z, y) \iff z$ lies between x and y .

Proof. If z lies between x and y , then $z = tx + (1 - t)y$ for some $t \in [0, 1]$, by the triangle inequality,

$$d(x, z) + d(z, y) = \|x - z\| + \|z - y\| \quad (6)$$

$$= \|x - tx + (t - 1)y\| + \|tx + (1 - t)y - y\| \quad (7)$$

$$= \|(1 - t)x - (1 - t)y\| + \|tx - ty\| \quad (8)$$

$$= (1 - t)\|x - y\| + t\|x - y\| = \|x - y\| \quad (9)$$

$$= d(x, y) \quad (10)$$

If we assume $d(x, y) = d(x, z) + d(z, y)$, then $\|x - y\| = \|x - z\| + \|z - y\|$, which implies that $x - z$ and $y - z$ are linearly dependent and $x - z = k(y - z)$ where $k \geq 0 \implies z = \frac{1}{1+k}x + \frac{k}{1+k}y$, and let $t = \frac{1}{1+k}$. Since $t \in [0, 1]$, we have $z = tx + (1 - t)y$ i.e. z lies between x and y . $\square$

LEMMA 1.7 (Young's Inequality). Let x, y be two non negative real numbers. Let $p > 1$ and q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (11)$$

Proof. We fix $y > 0$, and define the following function:

$$f(x) := \frac{x^p}{p} + \frac{y^p}{p} - xy \quad (12)$$

Applying the extrema test of single variable, we get that $f'(x_0) = 0$, and see that it holds for $x_0 = y^{1/(p-1)}$. Then we also see that $f''(y^{1/(p-1)}) > 0$, from which we can conclude that it is a minimum of the function on $(0, \infty)$, from which it follows that $f(x) \geq f(x_0)$, and here $f(x_0)$ happens to be zero. Therefore:

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q} \quad (13)$$

□

LEMMA 1.8 (Hölder's Inequality). *Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and for $1 \leq p \leq \infty$, we define $\|x\|_p = (\sum_{i=1}^n |x_i|^p)^{1/p}$ and for $p \rightarrow \infty$, we have $\|x\|_\infty = \max\{|x_i| : 1 \leq i \leq n\}$. For $p > 1$, let q be such that $\frac{1}{p} + \frac{1}{q} = 1$. Then the following inequality holds:*

$$\sum_i |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad \forall a, b \in \mathbb{K}^n \quad (14)$$

Proof. Let us take $x = \frac{|a_i|}{\|a\|_p}$ and $y = \frac{|b_i|}{\|b\|_q}$ and using Young's Inequality, we get:

$$\frac{|a_i| |b_i|}{\|a\|_p \|b\|_q} \leq \frac{1}{p} \left(\frac{|a_i|}{\|a\|_p} \right)^p + \frac{1}{q} \left(\frac{|b_i|}{\|b\|_q} \right)^q \quad (15)$$

Summing this over i from 1 to n , we get:

$$\frac{1}{\|a\|_p \|b\|_q} \sum_{i=1}^n |a_i| |b_i| \leq \frac{1}{p \|a\|_p^p} \sum_{i=1}^n |a_i|^p + \frac{1}{q \|b\|_q^q} \sum_{i=1}^n |b_i|^q \quad (16)$$

Simplifying this, we obtain:

$$\sum_{i=1}^n |a_i| |b_i| \leq \|a\|_p \|b\|_q \quad (17)$$

□

LEMMA 1.9 (Minkowski's Inequality). *Let $1 \leq p \leq \infty$, then we have Minkowski's inequality:*

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad \forall a, b \in \mathbb{K}^n \quad (18)$$

Proof. For $1 < p < \infty$, we observe that:

$$\|a + b\|_p = \left(\sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \quad (19)$$

$$\leq \left(\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} + \sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \right) \quad (20)$$

Now, we apply Hölder's inequality to both the terms, by taking $p = q(p - 1)$ and $1/q = (p - 1)/p$. Applying it to the left term:

$$\sum_{i=1}^n |a_i| |a_i + b_i|^{p-1} \leq \|a\|_p \left[\sum_{i=1}^n (|a_i + b_i|^{p-1})^q \right]^{1/q} \quad (21)$$

$$= \|a\|_p \left[\sum_{i=1}^n |a_i + b_i|^p \right]^{1/q} \quad (22)$$

$$= \|a\|_p \|a + b\|_p^{p/q} \quad (23)$$

Similarly for the right term, we have

$$\sum_{i=1}^n |b_i| |a_i + b_i|^{p-1} \leq \|b\|_p \|a + b\|_p^{p/q} \quad (24)$$

Using these two relations and simplifying, we finally obtain:

$$\|a + b\|_p \leq \|a\|_p + \|b\|_p \quad (25)$$

□

DEFINITION 1.10 (Open and Closed Balls). Let (X, d) be a metric space, and $x \in X$. Then for some $r > 0$, we define the open ball $B(x, r)$ and closed ball $B[x, r]$ as follows:

$$B(x, r) := \{y \in X \mid d(x, y) < r\} \quad (26)$$

$$B[x, r] := \{y \in X \mid d(x, y) \leq r\} \quad (27)$$

DEFINITION 1.11 (Line Segment). Let V be a real vector space and $x, y \in V$. We denote the line segment joining x and y by $[x, y]$, and we define to be $[x, y] := \{tx + (1 - t)y \mid 0 \leq t \leq 1\}$.

DEFINITION 1.12 (Convex Set). A subset $A \subset V$ of a vector space is said to be *convex* if $\forall x, y \in A$, the line segment $[x, y] \subset A$.

DEFINITION 1.13 (Open Set). For some $U \subset (X, d)$, the subset is said to be *open* if $\exists r > 0$ such that $B(x, r) \subset U \forall x \in U$.

LEMMA 1.14. Let (X, d) be a metric space, $x \in X$ and $r > 0$. Then the open ball $B(x, r)$ is an open set.

Proof. Let $y \in B(x, r)$. Assuming that $\exists s > 0$ such that $B(y, s) \subset B(x, r)$, if $z \in B(y, s)$, we want to show that $d(z, x) < r$. Now,

$$d(z, x) \leq d(z, y) + d(y, x) \leq s + d(x, y) < r$$

Therefore, we have $d(z, x) < r$, and thus $B(y, s) \subset B(x, r)$. Since y is arbitrary, we have proved the open ball $B(x, r)$ is an open set. □

DEFINITION 1.15 (Interval). If $I \subseteq \mathbb{R}$, then I is said to be an interval if and only if any $x, y \in I$ such that $x < y$ and $\forall z \in \mathbb{R}$ such that $x < z < y \implies z \in I$.

PROPOSITION 1.16. Any non empty subset $I \subseteq \mathbb{R}$ is an interval if it is of the form (a, b) , $[a, b]$, $(a, b]$, (a, b) , $(-\infty, \beta)$, (α, ∞) , $(-\infty, \beta]$ or $[\alpha, \infty)$, where $b < a$.

Proof. Dividing this into two cases: (1) bounded set and (2) unbounded set.

(1) Let $\sup I = b$ and $\inf I = a$. We claim that $(a, b) \subset I \subset [a, b]$. For $a < x < b$, then $x > a$ since a is the infimum, hence $\exists c \in I$ such that $c < x$. Similarly $x < b \implies \exists d \in I$ such that $x < d$, thus $c < x < d$ and it follows that $x \in I$ since its an interval. Since $x \in (a, b)$ is arbitrary, therefore $(a, b) \subset I$. As such, if we have $z \in I$ then $a \leq z \leq b$ by their definitions and hence $I \subset [a, b]$. Depending whether a and b actually are in I , we get (a, b) , $[a, b]$, $(a, b]$, $[a, b)$.

(2) For the unbounded case, let $b = \sup I$ and we claim $(-\infty, b) \subset I \subset (-\infty, b]$. Since b is a supremum, for some $x \in I$ we have $x < b$ and therefore $\exists d \in I$ such that $x < d$. We can also always find $c \in I$ such that $c < x$ and then we have $c < x < d \implies x \in I$. Therefore $(-\infty, b) \subset I \subset (-\infty, b]$, and we get the interval based on whether b is in I or not. Similarly, one can proceed with the (a, ∞) and $[a, \infty)$ cases. $\square$

LEMMA 1.17. A non empty open set in $\mathbb{R}$ is the union of countable family of pairwise disjoint open intervals.

Proof. Let $U \subseteq \mathbb{R}$ be an open set and $x \in U$. Now let I_x be the union of all sets such that $x \in I \subset U$, where I is an interval. Then, I_x would be the maximal interval containing x in U . Now, for $x, y \in U$, consider the intervals I_x and I_y . If they are not disjoint i.e. $I_x \cap I_y \neq \emptyset$, then their union would be a larger interval containing both, which would no longer be the maximal set unless $I_x = I_y$. So I_x and I_y must be either disjoint or identical. Then, we can write U can be written as $\bigcup_{i \in I} I_i$, where I is the index set.

Now, to prove countability, choose a rational $r_i \in I_i$ for each $i \in I$. Then, $\forall i \neq j \implies r_i \neq r_j$ since $I_i \cap I_j = \emptyset$, so we have a one to one map $i \mapsto r_i$ from I to $\mathbb{Q}$, hence I is countable. $\square$

THEOREM 1.18. Let (X, d) be a metric space and let $\mathcal{T}$ denote the set of all open sets in X , then:

1. $\emptyset, X \in \mathcal{T}$.
2. Arbitrary union of sets in $\mathcal{T}$ lies in $\mathcal{T}$.
3. Arbitrary intersection of sets in $\mathcal{T}$ lies in $\mathcal{T}$.

Proof.

1. $\emptyset$ is open by definition, and obviously X is open, so both are in $\mathcal{T}$.
2. Let $\{U_i \mid i \in I\}$ be a family of open sets. If U_k is open, then $\exists r > 0$ such that $B(x, r) \subset U_k \subset \bigcup_{i \in I} U_i$. Since x is arbitrary, it follows that the union of open sets is open in X , and it belongs to $\mathcal{T}$.
3. Let U_k be a finite collection of open sets in X . If the collection is pairwise disjoint, then $\bigcap_{k=1}^n U_k = \emptyset$ which is open in X . If not, let $x \in \bigcap_{k=1}^n U_k$, and since U_k is open then $\exists r_k > 0$ such that $B(x, r_k) \subset U_k$. Let $r := \min \{r_k \mid 1 \leq k \leq n\}$ and it follows that $B(x, r) \subset B(x, r_k) \subset U_k$ for each k i.e. $B(x, r) \subset \bigcap_{k=1}^n U_k$.

$\square$

DEFINITION 1.19 (Topological Space). Let X be a non empty set, then if $\mathcal{T}$ is a collection of subsets of X such that:

1. $\emptyset, X \in \mathcal{T}$.
2. An arbitrary union of sets in $\mathcal{T}$ lies in $\mathcal{T}$.
3. An arbitrary intersection of sets in $\mathcal{T}$ lies in $\mathcal{T}$.

Then $\mathcal{T}$ is said to be a *topology* of X and $(X, \mathcal{T})$ is called a *topological space*.

DEFINITION 1.20 (Equivalent Metrics). Two metrics d and d' are said to be *equivalent* if the topologies generated by the two are the same. This is equivalent to saying that $U \subset X$ is d -open if and only if it is d' -open.

DEFINITION 1.21 (Interior). Let $S \subset X$ be a subset of a metric space. If $x \in S$, it is said to be an *interior point* of S if $\exists r > 0$ such that $B(x, r) \subset S$. The set of interior points of S is denoted by S^0 and is called the *interior* of the set S .

If $(X, \mathcal{T})$ is a topological space, and $x \in A \subset X$ is said to be an interior point if $\exists$ an open set U with $x \in U$ such that $U \subset A$.

DEFINITION 1.22. Let $Y \subset X$ of a metric space. A subset $A \subset Y$ is said to open in Y if it is an open subset of metric space (Y, d) where d is the induced metric on Y .

DEFINITION 1.23 (Subspace Topology). If $Y \subset X$ is a subset of a topological space $(X, \mathcal{T})$, then the *subspace topology* $\mathcal{T}_Y$ is the collection $\{U \cap Y \mid U \in \mathcal{T}\}$.

DEFINITION 1.24 (Closed Set). A subset $F \subset X$ of a metric space is said to be *closed* if its complement $X - F$ is open in X .

THEOREM 1.25. Let $\mathcal{C}$ denote the class of closed sets in a metric space X , then:

1. $\emptyset, X \in \mathcal{C}$.
2. The intersection of an arbitrary family of closed sets is closed.
3. The union of a finite arbitrary family of closed sets is closed.

Proof.

1. $\emptyset$ and X are closed by definition.
2. Let $\{F_i \mid i \in I\}$ be a family of closed sets in $\mathcal{C}$. Since F_i is closed, its complement $U_i := X - F_i$ is open. From Theorem 1.18, we know the union of all such open sets is open and hence its complement, the intersection of all closed sets is closed.
3. Let $\{F_i\}$ be a finite set of closed subsets of X . Then the complement of F_i is U_i must be open, and from Theorem 1.18 we know a finite intersection of such sets is open. We take its complement to get that the finite union of closed is closed.

□